

CS305 Programming Assignment 2(2025F@sustech)

This assignment consists of Part 1 and Part 2, with a total score of 110 points. 100 points or above will be regarded as full marks, while the extra scores over 100 will **NOT** be added to other scoring items of CS305 course.

Part 1 - Lab Practices (50 pts)

- Complete 2 practices for each lab in the lab11, lab12, lab13, lab14 experimental slides. ($5 \times 2 \times 4 = 40$ pts)
- Practices on H3C device. (10 pts)
 - Log in to the H3C network device through the console port, show the summary information of all interfaces on the device, and distinguish between layer 2 interfaces and layer 3 interfaces. (2 pts)
 - Set up H3C devices remote login, allowing users to log in to the device via telnet on a local PC for viewing and configuration work. (4 pts)
 - Users can also log in to the device via telnet on non directly connected PCs for viewing and configuration work. (4 pt)

Tips

- No supplementary material submission needed for part 1
- You are expected to complete the practices check with the instructor/SAs in the experimental class before the practices' DDL.
- If you cannot complete them during the experiment class, please make an appointment with the instructor/SAs for a separate check and grading before the practices' DDL.
- The practices checked after DDL will NOT be counted towards the score of this assignment.

The DDL of practices for each lab is different, as follows:

- The DDL checked for the two practices in lab11 is 5:00 pm on Friday of week 13.
- The DDL checked for the two practices in lab12 is 5:00 pm on Friday of week 14.
- The DDL checked for the two practices in lab13 is 5:00 pm on Friday of week 15.
- The DDL checked for the two practices in lab14 and practices on H3C device is 5:00 pm on Friday of week 16.

Part 2 - Unicast, Multicast and ICMP(in IPv4/IPv6) (60 pts)

Introduction

Unicast communication follows a **one-to-one** model, where packets are sent from a single source to a single destination host, each identified by a unicast IP address. Most everyday Internet traffic, such as web browsing, email, or file transfers, relies on unicast. Each communication session between a client and a server is handled

independently, ensuring reliable delivery but potentially leading to redundant transmissions when multiple receivers request the same content.

In contrast, **Multicast** is a key technology at the network layer for implementing **one-to-many** service transmission. Its network architecture consists of multicast sources, multicast forwarding nodes (routers), and multicast group member nodes. The multicast source directs packets to the multicast group by specifying the destination as the **multicast group IP address** at the network layer and the **multicast group MAC address** at the link layer. Multicast group members can dynamically join or leave the group, enabling flexible allocation of network resources.

In this assignment, a hybrid virtual-real network topology based on eNSP is used as the experimental platform for both unicast and multicast validation. The code runs on the real network side, while the virtual network within the eNSP is used for verification testing. There are two tasks in this assignment:

- Task1. Complete the framework code(provided in the assignment) to implement the function of sending **ICMPv4/ICMPv6 Echo request** messages to the virtual network. While the destination is a unicast address, it initiate unicast communication, whereas while the destination is a multicast group address, it initiate multicast communication.
- Task2. Build a virtual network in eNSP and complete the unicast and multicast tests using the code from Task 1.

TIPS

In the tests, **Unicast** communication is similarly tested using ICMP or ICMPv6 messages(request and reply) to verify basic point-to-point connectivity between hosts. **Multicast** communication is verified by observing packet information, combined with the router's multicast forwarding table and forwarded packet statistics.

Goal and Grading Rule

IPv4 Unicast and Multicast (25 pts)

- **IPv4 Address Validation (5 pts):** For both **unicast** and **multicast** modes, your program needs to identify illegal IPv4 addresses and provide prompts during runtime. (**Note:** A valid multicast address **must** be within the range from **224.0.0.0** to **239.255.255.255**, and a valid unicast address **must** be within the range from **1.0.0.0** to **223.255.255.255**)
- **IPv4 Unicast Communication (10 pts):**
 - **Code:** You need to initiate the Unicast communication using **ICMP in IPv4** by sending **ICMP Echo Request** packets (The destination of its network layer is the unicast IPv4 addresss).
 - (Note: For ICMP header and checksum, you can refer [RFC 792 - Internet Control Message Protocol](#) & [IP报文格式大全 - 华为](#))
 1. Construct ICMP header (without checksum).
 2. Calculate the checksum based on the header and data of ICMP.
 3. Implement IPv4 unicast ICMP Echo request packets and send them by using **socket**.
 - **Test:** To test unicast communication, you should first build and confirm that the virtual network topology is correctly configured and that routing between subnets is established, then add cloud

devices and establish a connection between the virtual network and the real network through the cloud devices(Refer to Part B of the lab9 courseware).

- **IPv4 Multicast Communication (10 pts):**

- **Code:** You need to initiate the Multicast communication using **ICMP in IPv4** by sending **ICMP Echo Request** packets (The destination of its network layer is the multicast group IPv4 address, and the destination of its link layer is the multicast group MAC address).

(Note: For ICMP header and checksum, you can refer [RFC 792 - Internet Control Message Protocol](#) & [IP报文格式大全 - 华为](#))

1. Implement the conversion from IPv4 multicast address to MAC address (Refer to Page 11 of lab10 courseware).
 2. Construct ICMP header (without checksum).
 3. Calculate the checksum based on the header and data of ICMP.
 4. Implement IPv4 multicast ICMP Echo request packets and send them by using `socket`.
- **Test:** To test Multicast communication, you need to configure routers and PCs to support multicast routing and group management, then add cloud devices and establish a connection between the virtual network and the real network through the cloud devices(Refer to Part B of the lab9 courseware and practice 2 of lab10 courseware), your code works as the multicast source.

IPv6 Unicast and Multicast (25 pts)

- **IPv6 Address Validation (5 pts):** For both unicast and multicast modes, your program needs to identify illegal IPv6 addresses and provide prompts during runtime. (**Note:** A valid multicast address **must begin with** the binary `11111111`, which is hexadecimal `FF00::/8`, and a valid global unicast address **must be** within the range from `2000::` to `3FFF:FFFF:FFFF:FFFF:FFFF:FFFF:FFFF:FFFF`)

- **IPv6 Unicast Communication (10 pts):**

- **Code:** You need to initiate the Unicast communication using **ICMPv6 in IPv6** by sending **ICMPv6 Echo Request** packets (The destination of its network layer is the unicast IPv6 address).

(**Note:** For ICMPv6 header and checksum, you can refer [RFC 4443 - Internet Control Message Protocol \(ICMPv6\)](#) || [IP报文格式大全 - 华为](#) || [网络网络层之\(6\)ICMPv6协议](#))

1. Construct ICMPv6 header (without checksum).
 2. Calculate the checksum based on the header(IPv6 pseudo header + ICMPv6 header) and data of ICMPv6. For ICMPv6, checksum calculation should include IPv6 pseudo header.
 3. Implement IPv6 unicast ICMP Echo request packets and send them by using `socket`.
- **Test:** To test unicast communication, you should first build and confirm that the virtual network topology is correctly configured and that routing between subnets is established, then add cloud devices and establish a connection between the virtual network and the real network through the cloud devices(Refer to Part B of the lab9 courseware).

- **IPv6 Multicast Communication (10 pts):**

- **Code:** You need to initiate the Multicast communication using **ICMPv6 in IPv6** by sending **ICMPv6 Echo Request** packets (The destination of its network layer is the multicast group IPv6 address, and the destination of its link layer is the multicast group MAC address).

(**Note:** For ICMPv6 header and checksum, you can refer [RFC 4443 - Internet Control Message Protocol \(ICMPv6\)](#) || [IP报文格式大全 - 华为](#) || [网络网络层之\(6\)ICMPv6协议](#))
 1. Implement the conversion from IPv6 multicast address to MAC address (Refer to Page 30 of lab10 courseware).
 2. Construct ICMPv6 header (without checksum).
 3. Calculate the checksum based on the header(IPv6 pseudo header + ICMPv6 header) and data of ICMPv6. For ICMPv6, checksum calculation should include IPv6 pseudo header.
 4. Implement IPv6 unicast ICMP Echo request packets and send them by using **socket**.
- **Test:** To test Multicast communication, you should ensure that each device has valid IPv6 addressing and that multicast routing is properly configured. Unlike IPv4, the PC in eNSP does not support IPv6 multicast or MLD (Multicast Listener Discovery) operations. Therefore, in this topology, the router interface closest to the multicast member (assuming the PC represents a multicast receiver) is configured to simulate a multicast member(Refer to Part B of the lab9 courseware and practice 3 of lab10 courseware), your code works as the multicast source.

Report (10 pts)

NOTICE

If you complete the check for two practices in lab10 before 17:00 pm on Friday of week 12, you will receive a score of 10 in the report part regardless of whether you submit the report or not. Otherwise, you will need to write and submit the report according to the following requirements.

In the report, you need to present the following content:

NOTICE: DONOT use the same network topology in lab10 as the test topologies, as well as the unicast, multicast IPv4, and IPv6 addresses in the topology. Otherwise, the report will receive 0 points.

1. Screenshots of test topolog on IPv4 and IPv6 scenarios.

Missing any of the following items will result in a topology score of 0.

- (1)The network topology diagrams should clearly indicate each PC and Router interface's name and IP address, as well as the unicast address of multicast group members and the corresponding multicast group addresses.
 - (2)All devices and their interfaces must be labeled in the diagram for clarity.
2. Record Multicast Forwarding Table Changes Corresponding to Group Membership Updates in both IPv4 multicast and IPv6 multicast(use the command **display multicast forwarding-table** or **display multicast ipv6 forwarding-table**).

NOTICE: At least two different total number of multicast group members should be included.

- (1) Record incoming and outgoing interface in the Router's Multicast Forwarding Table Corresponding to Group Membership Updates in a table.
 - In the table(**referred to the table in practice 1 of lab 10**):
 - 1st column is the list of multicast group members, the 2nd and 3rd columns are the interface information in the multicast forwarding table of router near the multicast source: the 2nd column is the name and number of the incoming interface in the multicast forwarding table, and the 3rd column is the name and number of the outgoing interface in the multicast forwarding table.
 - Starting from the router closest to the multicast source and moving toward the routers near the multicast group members.
- (2) Screenshots of the multicast forwarding table on the router which is near the multicast source.
- 3. Record Data Packet Statistics Corresponding to Multicast Group Membership Changes in both IPv4 multicast and IPv6 multicast(use the command `display multicast forwarding-table` or `display multicast ipv6 forwarding-table`).

NOTICE: At least two different total number of multicast group members should be included.

- (1) Present the data packet statistics before and after multicast group membership changes in a table format, including:
 - a. The number of data packets sent from the multicast source.
 - b. The number of data packets received by each multicast group member node.
 - c. The number of multicast data packets forwarded by each multicast forwarding node (starting from the router closest to the Multicast source and moving toward the routers near the multicast group members). (use the command `display multicast forwarding-table`)
- (2) Screenshots are required for:
 - a. The number of data packets sent by the multicast source.
 - b. The number of data packets received by multicast group members.
 - c. Screenshots of the multicast forwarding table on the router which is near the multicast source, showing the forwarded data packet statistics, clearly indicating the packet forwarding counts.(use the command `display multicast forwarding-table`)

Implement and Test

In this part, you will use both Python and eNSP to implement and the testing of Unicast and Multicast. You will:

1. Use eNSP to set up and configure network topologies that support Unicast and Multicast communication.
2. Write a Python code to test Unicast and Multicast transmission using ICMP Echo Request packets.

You could refer to those methods to verify unicast and ICMP:

1. Packet Capture on Interfaces: Capture packets on the relevant interfaces and check:
 - (1) Whether there are ICMP echo request packets with valid unicast addresses as the destination.

- (2) If it is a valid ICMP echo request packet and is correctly sent to a connected virtual network, then the corresponding ICMP echo reply packet should be received, otherwise there is something wrong with the ICMP echo request packet or the hybrid virtual-real network.

You could refer to those methods to verify Multicast and ICMP:

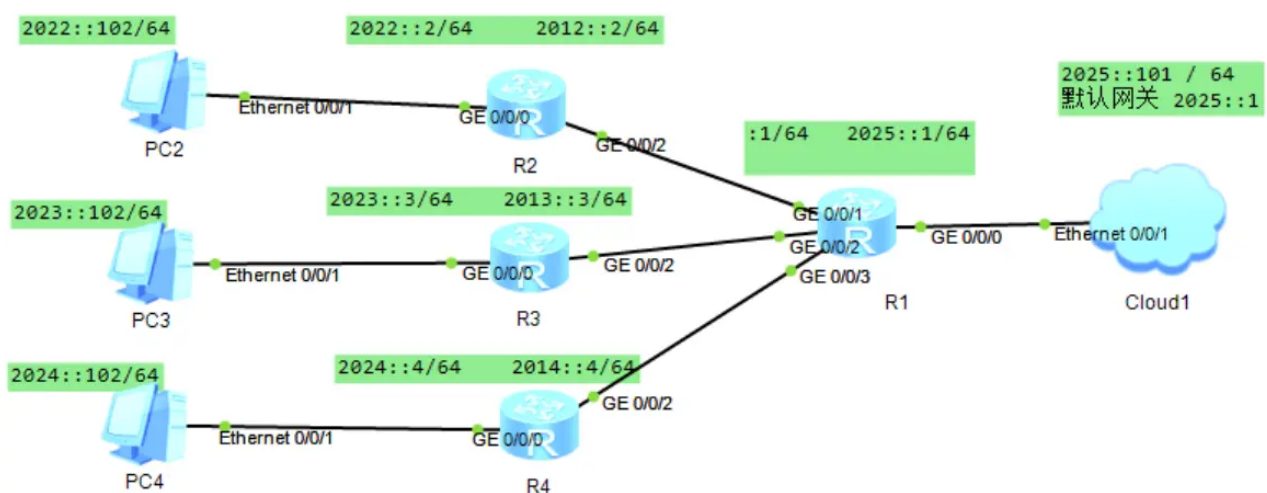
1. Packet Capture on Interfaces: Capture ICMP echo request packets on the relevant interfaces and check whether there are packets with valid multicast addresses as the destination.
2. On the Router: Use `display multicast forwarding-table` or `display multicast ipv6 forwarding-table` command to view the multicast forwarding table. This table shows the incoming and outgoing interfaces for multicast traffic on the local router, as well as the number of multicast packets matched and forwarded.
3. Note: The statistics in the multicast forwarding table are temporary and will age out over time. Therefore, it is important to run `display multicast forwarding-table` or `display multicast ipv6 forwarding-table` command immediately after sending multicast packets. If the interval is too long (about one minute or more), the statistics may disappear.

Code framework

The code framework in `cast_ping_simple.py` is for Part2 of this assignment. You are required to implement the logic inside the `TODO: Implement XXX` based on the assignment's requirements.

Notice:

1. **You are only allowed to use `raw_socket` for Unicast and Multicast implementation and transmission; Otherwise, you will receive 0 points.** This means the only permitted network-related library is `socket`. Other network-related libraries, such as `scapy`, are not allowed.
2. The multicast address's corresponding MAC address **MUST be generated internally within the program** (the mapping methods differ for IPv4 and IPv6).
3. **This assignment will be graded using scripts.** The following eNSP network topology is a example where your code will be tested in:



The testcases include IPv4 network and IPv4/IPv6 dual-stack network. The network topology may also be different.

4. For IP address validation, you only need implement and return **None** in validation functions if you find illegal IPv4 address, as the prompts have be provided in **run** function based on the return value from validation fuctions.
5. You may modify the entire code framework, but please make sure to keep the original parameter list, such as '-s'.
6. **The DDL for the part2 is before 5:00 pm on Friday of week 14.** Late submission is NOT allowed for part2.
7. It is recommended to use the router with the model name Route(not AR) in eNSP to build the network topology.

Submission

When submitting, you need to submit a .pdf report named **sid.pdf** and a python file containing all your code named **sid.py**. Package them into a .zip archive named **sid.zip** and submit it to blackboard site. **sid** is your student ID.

Your archive file tree should be like this:

```
sid.zip/  
├── sid.pdf  
└── sid.py
```

To facilitate script-based grading, please **make sure to** submit and name your files according to the specified format. Failure to do so may result in serious grading errors.

FAQ

Q: How do I connect the virtual network in eNSP and the real networks?

A: Please refer to the document titled "Implementing Bridging Between Routers and Network Cloud (Real PC Network Card) in eNSP"/ "eNSP 中实现路由器与网络云（真实PC 网卡）的桥接及IPv6 通信" for detailed instructions.

Q: eNSP Issue: Only UDP is available in the Cloud binding information.

A: The reason is that WinPcap was not installed correctly or was not completely uninstalled. You can see this [ENSP问题：Cloud绑定信息只有UDP一个](#) to solve it.